ABIHAIL NANS KUIPER

Data Science & AI/ML Engineer · Machine Learning · NLP · LLM Systems
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SUMMARY

ML engineer who ships: 10 end-to-end ML projects across computer vision, forecasting, NLP and tabular risk, plus secured, evaluated LLM systems. 3rd-year IT student (CGPA 9.2), DS & AI/ML Engineer Intern at Labmentrix, and IBM SkillsBuild × SkillUp Hackathon 2026 Regional Finalist.

EDUCATION

B.Tech, Information Technology — Kings Engineering College, Chennai *2024 – 2028*

- CGPA 9.2 / 10 (3rd year) · Data Structures, Neural Networks, Full Stack Development, Computer Networks, Cloud

SKILLS

Data Science & ML: Python, SQL, Pandas, NumPy, Scikit-learn, XGBoost, EDA, Feature Engineering, Statistics, Time-Series Forecasting, MLflow
Deep Learning & NLP: TensorFlow/Keras, CNNs, Transfer Learning, LSTM, Transformers, Hugging Face (DistilBERT, MuRIL, ViT)
LLMs & Agents: RAG, LangChain, LangGraph, CrewAI, FAISS, pgvector, Groq (LLaMA 3.3), Prompt Engineering
LLMOps & Evaluation: Guardrails AI, LangSmith, DeepEval, RAGAS, Portkey AI Gateway
BI & Deployment: Power BI, Tableau, Streamlit, Flask, FastAPI, Docker, Git, PostgreSQL, AWS (Foundations)

EXPERIENCE

DS & AI/ML Engineer Intern — Labmentrix, Bangalore (Remote) *May 2026 – Oct 2026*

- Built and deployed 10 ML apps in 5 months (computer vision, forecasting, recommenders, loan-risk ML, LLM agents), each with a public repo and a live demo (Streamlit Cloud, Gradio).
- Used leakage-safe validation (stratified and time-based splits) and took projects from data cleaning to live deployment.

Generative AI Intern — Prodigy InfoTech (Remote) *Mar 2026*

- Built a GPT-2 text generator showing how temperature and max_length change output quality, and a Stable Diffusion text-to-image generator. GitHub 1 · GitHub 2

KEY PROJECTS

Brain Tumor MRI Classification github.com/nansabi/Brain-tumor_MRI-Classification *Python · TensorFlow/Keras · Streamlit*

- 4-class classifier (glioma, meningioma, pituitary, none) on 2,443 cleaned images; custom residual + SE-block CNN rose from 57% to 83% after fixing the data pipeline.
- Fine-tuned 6 pretrained models; top-2 ensemble reached **95.12% test accuracy** (confusion matrices, per-class recall, F1, ROC); live Streamlit app.

EMI Predict AI — Loan Eligibility & EMI Prediction github.com/nansabi/EMI-Predict-AI *XGBoost · Scikit-learn · Streamlit*

- 400K+ records (IQR outlier handling, stratified split); 25 applicant inputs plus engineered ratio features via a ColumnTransformer pipeline; 6 models benchmarked.
- Checked that no feature was built from the target. XGBoost classifier **ROC-AUC 0.997** (vs 0.995 for Random Forest), F1-macro 0.84.
- Regressor RMSE ₹1,221 vs ₹3,553 for linear regression (66% lower); deployed on Streamlit Cloud.

PhonePe Pulse — Analytics & ML Forecasting github.com/nansabi/phonepe-pulse-dashboard *SQLite · Scikit-learn · Plotly*

- Turned 9,026 JSON files into a 47,254-row SQLite database (36 states); answered 12 business questions in SQL.
- Time-based split (trained on earlier years, tested on held-out 2023–24): **R^2 0.9966**, 2.87% average error. Finding: PhonePe grew 91× from 2018 to 2024.

SAFETEXT — Cyberbullying Detection github.com/nansabi *LinearSVC · DistilBERT · Groq LLaMA 3.3 · Flask*

- Hybrid 3-model ensemble on 47,692 tweets, 6 classes: TF-IDF + LinearSVC (**80.77% accuracy**) settles ~70% of inputs at ≥55% confidence; DistilBERT handles edge cases; LLaMA 3.3 gives the final label and explanation.
- Flask app with admin dashboard and human review, deployed on Railway.

AI Security Stack — Guarded & Evaluated RAG github.com/nansabi/AI-SecurityCourse *Guardrails AI · Portkey · LangSmith · DeepEval · RAGAS*

- Input guard blocks jailbreaks before the model is called → Portkey gateway (retry, fallback, caching) → RAG app → output guard validates the answer.
- Cached calls dropped from 3.4 s to 0.7 s; RAGAS faithfulness 0.92 (4-question sample); LangSmith traces each step.

Sentinel v2 — Self-Verifying Bug-Fixing AI github.com/nansabi/MultiAgent-Debugging-System *CrewAI · LangGraph · Docker · Groq/Gemini*

- Takes a public GitHub repo, clones, tests, diagnoses, fixes and proves the fix with mutation testing, on 3 orchestration engines.
- Docker sandbox with no network, human approval for every fix, full audit trail; presented to an IBM panel.

Also built: Vaccination analysis (SQL + Power BI) · Tourism recommender · Tesla LSTM forecaster · DeepFER (ViT, 71% on FER2013) · TamilBot (MuRIL, 91%+) · Multi-agent support · Travel agent · VidMind RAG · Deep Research Agent · n8n RAG pipeline · BI dashboards: Consumer Complaints · Netflix · Power BI Learning Journey

ACHIEVEMENTS

IBM Regional Hackathon Finalist — Team The HackOps *Sep 2026*

- Regional Finalist among 50+ teams, SkillUp × IBM SkillsBuild 2026 (R.V. College of Engineering); presented Sentinel to an IBM panel; Certificate of Excellence (22 Sep 2026).

CERTIFICATIONS & INVOLVEMENT

Certifications: Microsoft AI Fundamentals · AWS Cloud Foundations · Deloitte Technology Job Simulation (Forage) · MongoDB: Building RAG Apps, Vector Search Performance (Jul 2026) · IBM SkillsBuild: Ethical Considerations for Generative AI (Jul 2026) · AI Security Course
Involvement: IIC Coordinator (3+ events) · BlackOps Cybersecurity Club (8+ workshops) · SIH Internal Hackathon · 2 symposiums